Computer Programming
(1CS501CC26)

Gaming web portals.... To Learn while Playing

https://www.codedex.io/c-sharp
https://codecombat.com/teachers/guide/codecombat
https://www.codingame.com/ide/puzzle/there-is-no-spoon-episode-1

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01

SECTION 1

Foundations of Computing

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Foundations of Computing

Q1 **Why does a computer need a programming language?**

A. Computers understand only graphical interfaces
B. Computers ultimately execute machine-level instructions
C. Computers cannot perform calculations
D. Computers only understand English

*Answer: **B***

Q2 **Why is C called a middle-level language?**

A. It contains no high-level features
B. It supports high-level programming while also allowing low-level memory/hardware operations
C. It is halfway between Python and Java
D. It can only run on operating systems

*Answer: **B***

Q3 **A digital lock accepts a PIN only when two values are equal. Which operator is required?**

A. =
B. ==
C. !=
D. <=

*Answer: **B***

SECTION 1

Foundations of Computing

Q4 **Online banking lets you transfer money instantly. Which capability enables this?**

A. Local file storage only
B. Networked, real-time data processing and communication between systems
C. Only raw hardware speed
D. Assembly language

*Answer: **B***

Q5 **A pharma company simulates molecule interactions before physical trials. This is best described as:**

A. An application of computers in industry and scientific research
B. Entertainment
C. A web resource
D. Operating system design

*Answer: **A***

Q6 **Laptop advertised as "8 GB RAM, 512 GB SSD." Key functional difference?**

A. RAM is permanent, SSD is temporary
B. RAM is volatile working memory; SSD is non-volatile storage retaining data after power-off
C. Both are volatile
D. Both are non-volatile

*Answer: **B***

SECTION 1

Foundations of Computing

Q7 **Code: int x=5; printf("%d ",x++); printf("%d ",++x); — output?**

A. 5 6
B. 5 7
C. 6 6
D. 6 7

*Answer: **B***

Q8 **int x=10; float y=10.0; — which statement is most accurate?**

A. Both necessarily occupy the same memory
B. Both represent exactly the same data type
C. x and y represent the same numeric value but have different data types
D. x cannot be used in arithmetic

*Answer: **C***

Q9 **if(marks >= 40); printf("Pass"); else printf("Fail"); — key issue?**

A. >= is invalid
B. else cannot be used with if
C. The semicolon after if changes the intended control flow
D. marks must be float

*Answer: **C***

SECTION 1

Foundations of Computing

Q10 **ATM: allow withdrawal only when amount<=balance AND amount is a multiple of 100. Condition?**

A. amount<=balance || amount%100==0
B. amount>=balance && amount%100==0
C. amount<=balance && amount%100==0
D. amount<=balance && amount%100!=0

*Answer: **C***

Q11 **int a[4]={10,20,30,40}; for(i=0;i<=4;i++) printf("%d ",a[i]); — key problem?**

A. Arrays start from 1
B. i<=4 accesses one position beyond the valid index range
C. printf() cannot print array elements
D. The loop never executes

*Answer: **B***

Q12 **int a[5]={10,20}; — what are the remaining uninitialized elements?**

A. Garbage values
B. 0, 0, 0
C. 20, 20, 20
D. Compilation error

*Answer: **B***

SECTION 1

Foundations of Computing

Q13 **What is the role of a variable in C?**

A. Store a value that can be used and manipulated by a program
B. Store only programs permanently
C. Execute CPU operations directly
D. Display graphics

*Answer: **A***

Q14 **A traffic signal has states RED, YELLOW, GREEN. What type naturally represents these numerically?**

A. Character/string or integer code
B. Only floating point
C. Only array
D. Only pointer

*Answer: **A***

02

SECTION 2

Compilation, Memory & C Basics

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Q15 **For an array of 6 elements, how many swaps for a two-pointer reversal?**

A. 2
B. 3
C. 6
D. 4

*Answer: **B***

Q16 **Object file extension after assembly phase on Linux and Windows respectively?**

A. .o and .obj
B. .c and .h
C. .exe and .out
D. .s and .asm

*Answer: **A***

Q17 **What is the primary role of the linker in the C execution process?**

A. Loads the binary program from disk into RAM
B. Combines object files with library functions into a single executable
C. Checks C code for syntax errors
D. Converts C source directly into assembly

*Answer: **B***

SECTION 2

Compilation, Memory & C Basics

Q18 **Which memory section holds local variables defined inside a function?**

A. Data Segment
B. Text Segment
C. Heap Memory
D. Stack Memory

*Answer: **D***

Q19 **Correct format specifier to display the memory address of a variable?**

A. %a
B. %u
C. %d
D. %p

*Answer: **D***

Q20 **printf("Hello\rWorld"); — output?**

A. Hello\nWorld
B. World
C. Hello World
D. Hello\rWorld

*Answer: **B***

SECTION 2

Compilation, Memory & C Basics

Q21 **printf("%d", sizeof('A')); with sizeof(int)==4 — output?**

A. 4
B. 65
C. 1
D. 2

*Answer: **A***

Q22 **In CodeBlocks IDE, which shortcut compiles and runs simultaneously?**

A. F9
B. F10
C. F7
D. Ctrl + F9

*Answer: **A***

Q23 **int a = 012; printf("%d", a); — output?**

A. Compilation Error
B. 10
C. 012
D. 12

*Answer: **B***

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Compilation, Memory & C Basics

Q24 **Which header must be included to use malloc() and exit()?**

A. <stdlib.h>
B. <math.h>
C. <stdio.h>
D. <conio.h>

*Answer: **A***

Q25 **Which storage class gives a local variable persistent lifetime through program execution?**

A. extern
B. register
C. static
D. auto

*Answer: **C***

Q26 **printf("%%"); — output?**

A. %%
B. %
C. Undefined Behavior
D. Nothing

*Answer: **B***

SECTION 2

Compilation, Memory & C Basics

Q27 **What does a computer system bus do inside system architecture?**

A. Executes compiled C machine code
B. Converts source code into bytecode
C. Transmits data, signals, addresses between hardware components
D. Stores primary boot code permanently

*Answer: **C***

Q28 **Which type of memory is volatile and loses data when power is off?**

A. ROM
B. RAM
C. Flash EEPROM
D. HDD

*Answer: **B***

03

SECTION 3

Operators, Expressions & Control Flow

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Operators, Expressions & Control Flow

Q29 **Default initial value of an uninitialized local auto variable in C?**

A. NULL
B. -1
C. 0
D. Garbage value

*Answer: **D***

Q30 **What is the purpose of the #include directive in C?**

A. Links pre-compiled binary library files
B. Compiles individual functions into machine instructions
C. Inserts contents of the header file during preprocessing
D. Declares variable storage dynamically on stack

*Answer: **C***

Q31 **char x=127; x++; printf("%d",x); — output?**

A. -128
B. 128
C. Compilation Error
D. 0

*Answer: **A***

SECTION 3

Operators, Expressions & Control Flow

Q32

What does the const qualifier mean for a variable declaration?

A. Allocates storage permanently in global CPU registers
B. Ensures value is updated across threads
C. Makes variable visible only within current block
D. Prevents modifying the value after initial assignment

*Answer: **D***

Q33

int a=10,b=20; printf("%d", a+b>20); — output?

A. 0
B. 30
C. 20
D. 1

*Answer: **D***

Q34

Which statement correctly describes ASCII encoding used in C?

A. Assigns negative values to alphabetic characters
B. Uses 7-bit codes representing 128 unique character values
C. Uses 16-bit encoding for global unicode
D. Represents floats in scientific notation

*Answer: **B***

SECTION 3

Operators, Expressions & Control Flow

Q35 **Which statement describes a macro created via #define PI 3.14159?**

A. Textual substitution rule executed before compilation begins
B. Pointer referencing constant data segment
C. Typed floating point variable in heap memory
D. Function signature checked by compiler

*Answer: **A***

Q36 **What does the escape sequence \b do when printed in C?**

A. Moves cursor back one position on current line
B. Tabs cursor four spaces right
C. Inserts blank line before next character
D. Outputs an audible beep

*Answer: **A***

Q37 **Key structural difference between while and do-while loop?**

A. do-while does not require a semicolon
B. do-while evaluates condition after the body, guaranteeing one run
C. while accepts floating point conditions, do-while does not
D. while body always runs at least once

*Answer: **B***

SECTION 3

Operators, Expressions & Control Flow

Q38 **Evaluated output of the left shift bitwise operation 8 << 2?**

A. 64
B. 32
C. 2
D. 16

*Answer: **B***

Q39 **Which operator has the highest precedence among these?**

A. Binary Addition +
B. Relational Greater-Than >
C. Logical AND &&
D. Postfix increment ++

*Answer: **D***

Q40 **printf("A\bB"); — output?**

A. A
B. A B
C. AB
D. B

*Answer: **D***

SECTION 3

Operators, Expressions & Control Flow

Q41 **int x = 10 + 5*2/4 - 3%2; — evaluated value?**

A. 11
B. 10
C. 12
D. 9

*Answer: **A***

Q42 **int x=0,y=10; if(x!=0 && y/x>2) printf("True"); else printf("False"); — output?**

A. True
B. False
C. Division by Zero Runtime Error
D. Compilation Error

*Answer: **B***

04

SECTION 4

Loops & Arrays

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Q43 **Which ternary expression correctly finds the maximum of three numbers a, b, c?**

A. (a>b) ? (a>c?a:c) : (b>c?b:c)
B. (a>b && a>c) ? a : b : c
C. (a>b) ? a : (b>c) ? b : c (unparenthesized)
D. a>b>c ? a : c

Answer: ***A***

Q44 **In a switch-case statement, which data type is NOT allowed for the switch expression?**

A. int
B. char
C. enum
D. float

Answer: ***D***

Q45 **for(int i=0; i<10; i--); — what happens during execution?**

A. Loop executes 10 times
B. Loop does not execute
C. Infinite loop (or integer underflow loop)
D. Syntax error

Answer: ***C***

SECTION 4

Loops & Arrays

Q46 **Nested loop: for i=1..3 { for j=1..i { print j } } — output?**

A. 112123
B. 123
C. 12123
D. 112233

*Answer: **A***

Q47 **What happens if break executes inside an inner loop of a nested loop structure?**

A. Exits both inner and outer loops
B. Exits only the inner loop
C. Terminates the program
D. Restarts the inner loop

*Answer: **B***

Q48 **What is a "Pseudocode"?**

A. Machine code executed by the CPU
B. An informal high-level description of an algorithm for human reading
C. Compiled C code
D. A graphical flowchart tool

*Answer: **B***

SECTION 4

Loops & Arrays

Q49 **int x = 5 == 5; — result of evaluating this in C?**

A. x = 5
B. x = 1
C. x = 0
D. Syntax Error

*Answer: **B***

Q50 **printf("%d", 10 || 0); — output?**

A. 10
B. 0
C. 1
D. True

*Answer: **C***

Q51 **int arr[3]={5,10,15}; printf("%d", arr[3]); — output?**

A. 15
B. 0
C. Garbage value or Segmentation Fault
D. Compilation Error

*Answer: **C***

SECTION 4

Loops & Arrays

Q52 **int a[3]={1,2,3}; int b[3]; b = a; — what happens?**

A. Copies elements of a into b
B. Copies the base address of a into b
C. Compilation Error
D. Runtime Error

*Answer: **C***

Q53 **How many total elements in a 2D array declared as int matrix[3][4]?**

A. 7
B. 12
C. 3
D. 4

*Answer: **B***

Q54 **Which memory layout order do standard C compilers use for 2D arrays?**

A. Column-Major Order
B. Row-Major Order
C. Diagonal Order
D. Random Block Order

*Answer: **B***

SECTION 4

Loops & Arrays

Q55

How does C pass arrays to functions by default?

A. Pass by Value (copies complete array)
B. Pass by Reference / base address decay
C. Pass by Name
D. Pass by File Handle

*Answer: **B***

Q56

Where are Variable-Length Arrays (VLAs) allocated in memory?

A. Heap Memory
B. Stack Memory
C. Global Data Segment
D. Code/Text Segment

*Answer: **B***

05

SECTION 5

Practical Programming Tasks

APPLIED PROBLEM

Practical 1 : Find the Bug!

PROBLEM

```
int marks = 75;
if(marks > 40);
{ printf("Pass"); }
else
{ printf("Fail"); }
```

SOLUTION / KEY LOGIC

✓ **Semicolon after if(marks > 40);**

The semicolon terminates the if statement immediately, making the { printf("Pass"); } block independent and causing an 'else without a previous if' error.

APPLIED PROBLEM

Practical 2: ATM Currency Note Dispenser Logic

PROBLEM

Dispense Rs. 387 using notes of Rs.100, 50, 20, 10.

SOLUTION / KEY LOGIC

Number of Rs.100 notes = 387 / 100 → 3 notes

Remaining balance = 387 % 100 → Rs. 87

APPLIED PROBLEM

Practical 3: Crack the Digital Lock (Number Systems)

PROBLEM

Digital lock stores password in binary: 101101

SOLUTION / KEY LOGIC

✓ **Binary 101101 → Decimal: 1*32+0*16+1*8+1*4+0*2+1*1 = 45**

45 (Decimal) = 2D (Hexadecimal) — binary, decimal and hex represent the same value in different bases.

APPLIED PROBLEM

Practical 4: Food Delivery Price Analyzer

PROBLEM

Prices: 120, 250, 90, 450, 180, 300, 150, 80, 500, 220

SOLUTION / KEY LOGIC

Find: Max, Min, Average, items > ₹ 200, items < ₹ 100, total value

Bonus: apply a 20% discount to items priced strictly above ₹ 300 and display the updated price list.

APPLIED PROBLEM

Practical 5: Digital Base Converter

PROBLEM

Build an interactive console program to convert between number bases.

SOLUTION / KEY LOGIC

✓ **Decimal ↔ Binary, Decimal ↔ Hexadecimal, Binary ↔ Hexadecimal**

A menu-driven program offering all six conversion directions.

APPLIED PROBLEM

Practical 6: Game Score Analyzer

PROBLEM

Scores: 25, 40, 10, 80, 65, 90, 35, 55, 70, 100

SOLUTION / KEY LOGIC

Find: highest/lowest score, average, best round, rounds above average, rounds below 50

Also reverse the score array in-place and print the latest round's score first.

APPLIED PROBLEM

Practical 7: Security Code & OTP Pattern Analyzer

PROBLEM

Input a numeric security code, e.g. 47492741, into an integer array.

SOLUTION / KEY LOGIC

Find repeated digits, frequency of each digit, most frequent digit

Also identify the first repeated digit in the sequence.

APPLIED PROBLEM

Practical 8: Number System "Escape Room"

PROBLEM

Encrypted hex ASCII string: 66 6C 61 67

SOLUTION / KEY LOGIC

66→102→'f' 6C→108→'l' 61→97→'a' 67→103→'g'

Decoded output: "flag"